

Aditi Prabhu A

Software Development Engineer
Backend, AI Agents & Cloud Infrastructure
Bangalore, India

+91-8073370211
✉ aditiprbh@gmail.com
🐙 GitHub Profile
🌐 LinkedIn Profile

SUMMARY

Founding engineer at a fintech startup, building the backend platforms, AI-agent systems, and AWS infrastructure that run financial operations products. Experienced in owning the stack from API and PostgreSQL design through long-running agent workflows, CI/CD, and production deployment.

EDUCATION

- **PES University** 2021-25
Bachelor of Technology in Computer Science and Engineering

EXPERIENCE

- **OneCap Technologies Private Limited** Jan 2025 – Present
Software Development Engineer I Bangalore
 - Architected and deployed 8 extensible backend services using Python, FastAPI, and TypeScript on Amazon ECS Fargate, Lambda, ALB, and S3, covering reconciliation, data ingestion, and internal platforms.
 - Designed and owned PostgreSQL architecture across platform services, including schema design, FDW-based cross-service data access, indexing strategies, and query optimization for high-volume reconciliation workloads.
 - Automated long-running reconciliation workflows using AWS Step Functions, reducing manual operational effort through resilient retry, polling, and failure-handling mechanisms.
 - Built a long-running reconciliation agent with the Claude Agent SDK and Amazon Bedrock AgentCore capable of processing approximately 200,000 transactions.
 - Reduced Docker CI/CD build times by ~80% through dependency-layer caching and build optimization, accelerating deployment velocity across all backend services.
 - Set up virtual engineering agents on AWS Lightsail and orchestrated Discord and Google Chat collaboration workflows for debugging and engineering tasks, backed by a live, self-maintaining knowledge base of the codebase.
- **Center for Cloud Computing and Big Data** June – September 2024
Summer Research Intern Bangalore
 - Built distributed data processing pipelines using PySpark and Hadoop for large-scale anomaly detection across edge network nodes, achieving 99% detection accuracy.
 - Designed automated remediation workflows integrating MQTT-based IoT infrastructure and CI/CD pipelines, significantly reducing operational response time.

PROJECTS

- **Agent Realm** Python, FastAPI, PostgreSQL, AWS, MCP
Full-stack web and MCP platform for managing always-on AI agents
 - Built Checkpoint end-to-end: a FastAPI and PostgreSQL backend on AWS (Lambda, ALB, Aurora, Cognito, CDK) with a dashboard where operators set goals and review evidenced agent check-ins.
 - Added an MCP server so agents authenticate with hashed API keys, fetch active goals, file check-ins, and pick up feedback through the same domain logic as the dashboard, keeping one performance trail per agent.
- **Lucid Care** Python, PyTorch, Graph RAG
Multi-stage ML pipeline system for large-scale clinical data processing
 - Built a multi-stage ML pipeline processing large-scale clinical datasets for prediction and recommendation workloads, achieving 96% model accuracy.
 - Integrated knowledge graphs and Graph RAG pipelines to improve contextual reasoning and reduce false positives in model outputs.

TECHNICAL SKILLS

Languages: Python, TypeScript

Cloud: AWS (ECS Fargate, Lambda, ALB, S3, Step Functions, IAM, VPC, SES, CloudWatch, ECR, API Gateway, Lightsail, Aurora RDS, Cognito, EventBridge)

Databases: PostgreSQL, MongoDB

AI / Agents: Amazon Bedrock, AgentCore, Claude Agent SDK, OpenClaw, MCP

Delivery: Docker, GitHub Actions, CI/CD, Pytest, Git, Linux

Big Data: PySpark, Hadoop